

THE HONG KONG ASSOCIATION FOR COMPUTER EDUCATION  
INFORMATION AND COMMUNICATION TECHNOLOGY MOCK EXAM 2026

**PAPER 1 (Section A)**

Time Allowed: 8:30 am – 10:30 am

(2 hours for both Section A and B)

This paper must be answered in English.

**GENERAL INSTRUCTIONS**

1. There are TWO sections, A and B, in this Paper.
  2. Section A consists of multiple-choice questions in this question book. Section B contains conventional questions printed separately in Question-Answer Book.
  3. Answers to Section A should be marked on the Multiple-choice Answer Sheet. Answers to Section B should be written in the spaces provided in the Question-Answer Book. **The Answer Sheet for Section A and the Question-Answer Book for Section B must be handed in separately at the end of the examination.**
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**SECTION A (MULTIPLE-CHOICE QUESTIONS)**

**INSTRUCTIONS FOR SECTION A**

1. Read carefully the instructions on the Answer Sheet. After the announcement of the start of the examination, you should first stick a barcode label and insert the information required in the spaces provided. No extra time will be given for sticking on the barcode label after the 'Time is up' announcement.
2. When told to open this book, you should check that all the questions are there. Look for the words '**END OF SECTION A**' after the last question.
3. All questions carry equal marks.
4. **ANSWER ALL QUESTIONS.** You are advised to use an HB pencil to mark all the answers on the Answer Sheet, so that wrong marks can be completely erased with a clean rubber. You must mark the answers clearly; otherwise you will lose marks if the answers cannot be captured.
5. You should mark only **ONE** answer for each question. If you mark more than one answer, you will receive **NO MARKS** for that question.
6. No marks will be deducted for wrong answers.

**There are 40 questions in this paper. Choose the most suitable answers.**

1. Mary is helping to input the names of the winners on the Sports Day. Her teacher invited another student, Carrie, to do the same input as Mary. What error checking method is being used?

- A. Type check
- B. Validity check
- C. Double data Entry
- D. Parity check

2. Which pair of the file format and data type is correct?

| <u>Format</u> | <u>Data type</u> |
|---------------|------------------|
| A. DOCX       | audio            |
| B. SVG        | image            |
| C. GIF        | text             |
| D. MP3        | video            |

3. Mr. Wong is the manager of a large chain electronics store. He is designing a program for inventory record registration to facilitate staff in managing product information. Which of the following sets of data types is most appropriate for this program?

| <u>Product ID</u> | <u>Stock Quantity</u> | <u>Unit Price</u> | <u>Supplier</u> |
|-------------------|-----------------------|-------------------|-----------------|
| A. Float          | Float                 | Integer           | String          |
| B. Integer        | Integer               | Float             | String          |
| C. String         | Integer               | Float             | String          |
| D. String         | Float                 | Integer           | String          |

4. A computer received four data sequences, each containing a parity bit. Only one sequence was corrupted during transmission. Which sequence is corrupted?

- A. 1111 0011
- B. 0001 1001
- C. 1100 1011
- D. 1000 0000

Study the following situation and answer Questions 5 and 6.

Mario found an old handheld game console in a vintage shop. He knew that each pixel is represented by 2 bytes and the bit pattern of each pixel is `NR``RR``RR``RR` `RR``GG``GG``GG` `GG``GB``BB``BB` where the first bit `N` is always `0` and the bits `RRRRR`, `GGGGG` and `BBBBB` represent the magnitude of red, green and blue component of a pixel respectively.

5. How many different colors can the palette of the device support?

- A. 255
- B. 1,024
- C. 16,384
- D. 32,768

6. Which of the following bit patterns in hexadecimal represents a pure colour of red, green or blue?

- (1) `003F`
- (2) `03E0`
- (3) `7C00`

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

7. Which of the following processes require(s) conversion between analog and digital signals?
- (1) Scanning a physical photograph to a computer
  - (2) Compressing a folder of files into a single ZIP file
  - (3) Playing a song from a music streaming service through headphones
- A. (1) only  
B. (2) only  
C. (1) and (3) only  
D. (2) and (3) only
8. A company is designing a database to store customer names from around the world, including names written in English, Arabic (e.g., محمد), and Korean (e.g., 박서준). Which of the following character sets must be used to ensure all names are stored correctly?
- (1) UTF-8
  - (2) ASCII
  - (3) BIG-5
- A. (1) only  
B. (2) only  
C. (1) and (3) only  
D. (2) and (3) only
9. Assume every person in Hong Kong has a unique Hong Kong Identity Card (HKID) number. In consideration of privacy, which of the following databases would the HKID number **not** be a suitable primary key?
- A. Hospital patient records
  - B. Public library membership database
  - C. A company's customer purchase history
  - D. Voter registration database

10. The following two worksheets, SalesData and Summary, are used to track product sales.

SalesData

|     | A                  | B             | C              | D                 |
|-----|--------------------|---------------|----------------|-------------------|
| 1   | <b>Salesperson</b> | <b>Region</b> | <b>Product</b> | <b>Units Sold</b> |
| 2   | John               | North         | Widget A       | 150               |
| 3   | Jane               | South         | Widget B       | 200               |
| 4   | Peter              | North         | Widget C       | 120               |
| ... | ...                | ...           | ...            | ...               |
| 500 | Susan              | West          | Widget A       | 80                |

Summary

|   | A             | B                  |
|---|---------------|--------------------|
| 1 | <b>Region</b> | <b>Total Units</b> |
| 2 | North         |                    |
| 3 | South         |                    |
| 4 | East          |                    |

A formula is entered into cell B2 of the Summary worksheet by the sales manager to calculate the total units sold for the region listed in A2. This formula is then copied down to cells B3 and B4 to find the totals for the other regions. Which of the following is the correct formula for cell B2?

- A. =SUMIF(SalesData!B2:B500, A2, SalesData!D2:D500)
  - B. =SUMIF(SalesData!\$B2:\$B500, A2, SalesData!\$D2:\$D500)
  - C. =SUMIF(SalesData!\$B\$2:\$B\$500, A2, SalesData!\$D\$2:\$D\$500)
  - D. =SUMIF(SalesData!\$A\$2:\$A\$500, A2, SalesData!\$D\$2:\$D\$500)
11. Lily needs to write an SQL query to retrieve records from a table for students with heights greater than 160. The results must be sorted by height from shortest to tallest. Which of the following code segments is correct?
- A. WHERE height > 160 ORDER BY height ASC
  - B. HAVING height >= 160 ORDER BY height DESC
  - C. WHERE height < 160 ORDER BY height ASC
  - D. HAVING height > 160 ORDER BY height

12. Regarding the relationship between memory components and performance, which of the following is/are correct?
- (1) A larger RAM size generally improves performance by reducing the need for accessing secondary storage.
  - (2) A wider memory address bus allows the computer to access a larger amount of physical memory.
  - (3) An increase in word length (e.g., from 32-bit to 64-bit) typically decreases data transfer speed.
- A. (1) only  
B. (3) only  
C. (1) and (2) only  
D. (2) and (3) only
13. Which of the following best explains why increasing the size of RAM doesn't always improve performance?
- A. RAM is slower than ROM.  
B. Software may not utilize additional RAM.  
C. CPU frequency limits RAM usage.  
D. RAM size affects only graphics performance.
14. Which of the following is/are the common specification(s) of a VR headset?
- (1) 802.11ax
  - (2) 8000 mAh
  - (3) 512 GB
- A. (1) and (2) only  
B. (2) and (3) only  
C. (1) and (3) only  
D. (1), (2) and (3)

15. Which of the following statements about CPU architecture is true?
- A. Word length affects instruction complexity.
  - B. Clock rate determines instruction set size.
  - C. ALU is responsible for decoding instructions.
  - D. CU performs logical operations.
16. In computer memory hierarchy, which type of memory has highest data transfer rate but has the smallest capacity?
- A. Hard disk
  - B. Solid-state disk
  - C. RAM
  - D. CPU cache
17. Regarding Internet Protocol (IP) addresses, which of the following statements are true?
- (1) IPv4 addresses are longer and offer more unique addresses than IPv6.
  - (2) IPv6 was developed to solve the problem of running out of IPv4 addresses.
  - (3) Both IPv4 and IPv6 are used to identify devices on a network.
- A. (1) and (2) only
  - B. (1) and (3) only
  - C. (2) and (3) only
  - D. (1), (2) and (3)
18. Why are fibre optic cables preferred for long-distance communication?
- A. They are cheaper than UTP cables.
  - B. They have minimal signal loss and high bandwidth.
  - C. They are easier to install than microwave links.
  - D. They are compatible with all wireless standards.

19. HTTPS is used to browse a secure online banking system. What role(s) does it play?
- (1) Encrypt data for security
  - (2) Divide and reassembles packets
  - (3) Translate URLs into IP addresses for server access
- A. (1) only  
B. (3) only  
C. (1) and (2) only  
D. (2) and (3) only
20. What is the primary purpose of a network switch?
- A. To connect multiple networks together.  
B. To forward data packets within a LAN.  
C. To convert digital signals to analog signals.  
D. To assign IP addresses to devices.
21. A user reports that his Internet connection is sometimes very slow, especially when many devices are connected. Which factors might contribute to these issues?
- (1) Limited bandwidth from the Internet Service Provider.
  - (2) High network traffic or congestion on the local Wi-Fi.
  - (3) Interference from neighboring wireless networks.
- A. (1) only  
B. (2) only  
C. (1) and (2) only  
D. (2) and (3) only



22. If your web browser warns you about an "untrusted connection" or "certificate error" when visiting an HTTPS website, what might be the problem?
- (1) The website's security certificate has expired or is invalid.
  - (2) Your computer's DNS settings are incorrect.
  - (3) The website's identity cannot be verified by your browser.
- A. (1) and (2) only  
B. (1) and (3) only  
C. (2) and (3) only  
D. (1), (2) and (3)
23. Arrange the following addresses in the descending order of number of bits: IPv4, IPv6, MAC
- A. IPv4 > MAC > IPv6  
B. IPv4 > IPv6 > MAC  
C. IPv6 > IPv4 > MAC  
D. IPv6 > MAC > IPv4
24. How does the Domain Name System (DNS) use caching to improve Internet performance?
- (1) It stores frequently accessed domain name-to-IP address translations locally.
  - (2) It encrypts DNS queries to speed up resolution times.
  - (3) It makes websites load faster by directly serving cached web content.
- A. (1) only  
B. (2) only  
C. (1) and (3) only  
D. (2) and (3) only

25. What does a phishing attack typically involve?

- A. Overloading a server with traffic
- B. Encrypting files and demanding ransom
- C. Tricking users into revealing sensitive information
- D. Stealing hardware devices

26. What is the output of the following algorithm?

```
X ← 5
Y ← 1
WHILE X > 0 DO
    Y ← Y * 2
    X ← X - 1
OUTPUT Y
```

- A. 16
- B. 24
- C. 32
- D. 64

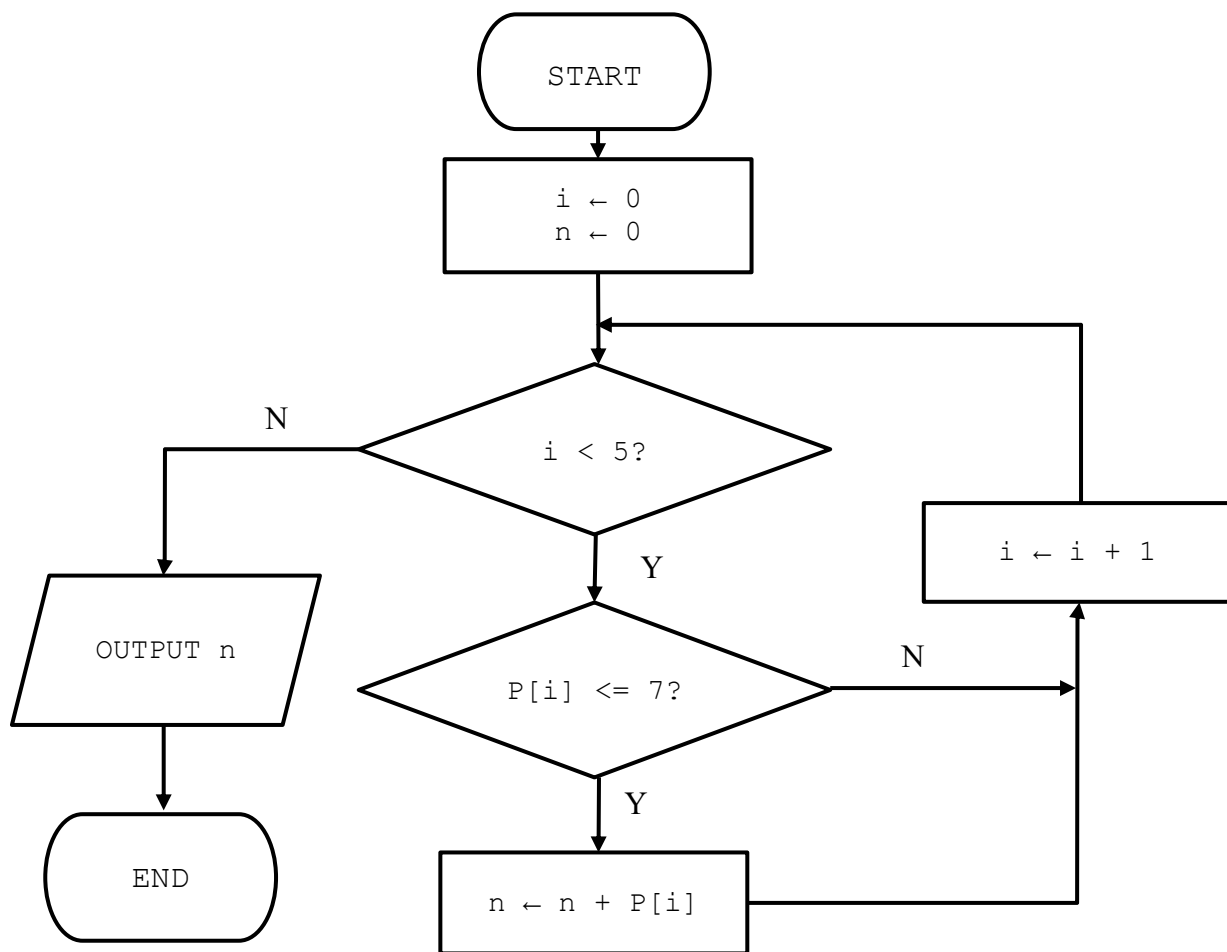
27. What is the output of the following algorithm?

```
B ← 1
FOR i FROM 1 TO 4 DO
    IF i > 2 THEN
        OUTPUT B
        B ← B + 3
    ELSE
        OUTPUT i
```

- A. 1 2 1 4
- B. 1 2 3 6
- C. 1 2 4 7
- D. 1 2 5 8

Study the following flowchart and answer Questions 28 and 29.

The initial values of  $P[0]$  ,  $P[1]$  ,  $P[2]$  ,  $P[3]$  and  $P[4]$  are 10, 8, 6, 7 and 5 respectively.



28. What is the output value of  $n$ ?

- A. 4
- B. 18
- C. 26
- D. 36

29. What is the purpose of the program?

- A. Calculate the sum of the values in the array.
- B. Calculate the number of values and their sum in the array.
- C. Calculate the sum of values less than or equal to 7 in the array.
- D. Calculate the number of values and the sum of values less than or equal to 7 in the array.

30. Study the following algorithm:

```
A ← 1
B ← 1
REPEAT
    OUTPUT B
    C ← A + B
    A ← B
    B ← C
UNTIL (B > 15)
```

How many times will 'OUTPUT B' be executed?

- A. 5
- B. 6
- C. 7
- D. 8

31. The following algorithm is used to find the value of  $(1^3 + 2^3 + 3^3 + \dots + N^3)$ .

```
INPUT N
S ← 0
WHILE  DO
    S ← S + N * N * N
    N ← N - 1
OUTPUT S
```

What should be in the box?

- A.  $N > 0$
- B.  $N > 1$
- C.  $N = 0$
- D.  $S > 0$

32. Suppose that all values in the array N are 3. What is the output of algorithm below?

```
S ← 0
i ← 0
WHILE (i ≤ 4) DO
    S ← S + N[i]
OUTPUT S
```

- A. 9
- B. 12
- C. 15
- D. Cannot be found.

33. Y is an array. What is the output of the following algorithm?

```
Y[1] ← 5
FOR k FROM 1 TO 6 DO
    Y[k+1] ← Y[k] * k
OUTPUT Y[7]
```

- A. 120
- B. 600
- C. 3000
- D. 3600

34. What are the boundary cases for testing in the following segment of an algorithm?

```
INPUT A
IF A > 8 THEN
    B ← 15
ELSE
    B ← 25
OUTPUT A, B
```

- (1) 8
- (2) 9
- (3) 15
- (4) 25

- A. (1) and (2) only
- B. (3) and (4) only
- C. (2), (3) and (4) only
- D. (1), (2), (3) and (4)

35. Suppose `score` is an integer variable and `isComplete` is a Boolean variable. Which of the following Boolean expressions produce the same result?

- (1) `(score >= 60) OR (isComplete = TRUE)`
- (2) `(score < 60) AND (isComplete = TRUE)`
- (3) `NOT ((score < 60) AND (isComplete = FALSE))`

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

36. Mr. Lee is an IT manager at a charity organization. He wrote the following algorithm. Array A is used to record the gross income of each event. According to regulations, 30% of the gross income will be allocated to basic operating costs. The following algorithm outputs the total amount of these costs.

```
A ← [31000, 42000, 55000, 21500, 18750]
Total ← 0
Cost ← 0
FOR i FROM 1 TO Length of A
    
Cost ← Total * 30%
OUTPUT Cost
```

- A. Cost ← Cost + A[i]
- B. Total ← Total + i
- C. Cost ← Cost + i
- D. Total ← Total + A[i]

37. What are the suitable testing values for the following segment of an algorithm?

```
IF (NUMBER > 0) THEN
    OUTPUT 'POSITIVE'
ELSE IF (NUMBER < 0) THEN
    OUTPUT 'NEGATIVE'
ELSE
    OUTPUT 'ZERO'
```

- A. -5, 10, 20
- B. -1, 0, 5
- C. 0, 2, 3
- D. -10, -20, -30

38. Which of the following attributes for a product on an e-commerce website can be defined as a Boolean data type?
- (1) Product name
  - (2) Inventory availability status
  - (3) Price
- A. (1) only  
B. (2) only  
C. (1) and (3) only  
D. (2) and (3) only
39. Which of the following actions can promote equitable access for youth?
- A. Shifting paperwork to digital formats
  - B. Requiring everyone to use artificial intelligence in learning and working
  - C. Providing scholarships for outstanding students with IT skills
  - D. Offering free IT lessons with necessary equipment
40. David learnt to use artificial intelligence platforms to generate images and program codes. He created a multi-player game and published it as freemium software with a monthly subscription after a 2-week trial period. Which of the following potential risks does he face?
- A. Copyright issues, as he did not check the copyrights of the images.
  - B. Patent infringement, as he does not own the algorithm of the program.
  - C. Trademark issues, as he does not share the revenue with the artificial intelligence platforms.
  - D. Money laundering, as he charges users for freemium software.

**END OF SECTION A**